MATH 101 Unit 1: Expressions, Equations, and Inequalities

Practice 1 | Version 202610122020.34 | 30 minutes

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

QUESTIONS	ANSWERS
1. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 3$	1. Derived bounds 12 pts Interval 8 pts
2. Simplify and state all values excluded by the original expression. $\frac{x^2-4}{x^2+2x-8}$	2. Expression 14 pts Excluded values 6 pts
3. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-2}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{18}$	3. (a) 10 pts (b) 10 pts
4. Solve both equations. In (b), state original restrictions and check your answer. $(a)\quad x^2+2x-3=0$ $(b)\quad \frac{4}{2x-4}+\frac{1}{x-2}=3$	4. (a) 10 pts (b), with check 10 pts
5. (a) Expand and simplify. (b) Factor completely. $(a)\quad (2x-3)(x+2)$ $(b)\quad 2x^2-5x-3$	5. (a) 10 pts (b) 10 pts

Instructor answer key: MATH 101 Unit 1: Expressions, Equations, and Inequalities

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. Derived bounds

$-1 \le x \le 2$

12 pts

Interval

$[-1, 2]$

8 pts

2. Expression

$\frac{x+2}{x+4}$

14 pts

Excluded values

$\{-4, 2\}$

6 pts

3. (a)

$\frac{2y^4}{x^3}$

10 pts

(b)

$3\sqrt{2}$

10 pts

4. (a)

$x = -3,\ 1$

10 pts

(b), with check

$x = 3;\ x \neq 2$

10 pts

5. (a)

$2x^2 + x - 6$

10 pts

(b)

$(x-3)(2x+1)$

10 pts